



Vitro diagnosis nanodiamonds

Study shows that quantum sensing abilities of nanodiamonds can help detect disease earlier. <https://dgit.in/dec20-37>



Spirula Sighting

Marine biologists have captured the first live footage of the very elusive, deep-sea ram's horn squid. <https://dgit.in/dec20-38>

sands at time. First it will propose a mapping for the letters using a basic frequency analysis. For example, it might find the most common letter of the unknown language, and map it to the letter “E”. Once the most frequent alphabets in both the languages are mapped, it will scan the text to find which of the probable mappings can be used to identify prefixes and suffixes. At this time, the system is also looking for correspondences between words, which in turn refine the correspondences between alphabets. The system then keeps improving the guesses, by mapping known alphabets and words to unknown alphabets and words. Finally, the system reaches a stage where the successive guesses are not increasingly probable to be true, and at that point there is a consistent set of mappings. This means that the system has solved the unknown language.

To gauge the accuracy of the system, the researchers used it to decipher Ugaritic, using the related language Hebrew, for the purposes of the study. The task for the computer was to find the unknown Ugaritic letters and words, and map them to letters and words in the Hebrew language. Ugaritic was a language discovered in modern day Syria by French archaeologists in 1929. The language has already been deciphered by humans, but the researchers from MIT made the computers do it anyway, so that they could verify how accurate the system was. Out of the 30 alphabets in the Ugaritic language, the system managed to correctly map 29 to Hebrew letters. About one third of all Ugaritic words have cognates in Hebrew, and the system managed to identify 60 percent of these cognates, with the spellings being off by only a single character in most of the remaining “wrong” guesses. One of the promises of the technology was that it could help improve the accuracy of translating between known languages, and allow companies such as Google to offer translations between more languages, faster.

There were a number of shortcomings identified in the system though. The system cannot derive the meanings of words from the context, which is something humans can do easily. For example, in Ugaritic the words for “daughter” and “house” are very similar, but that is not the case in Hebrew. The computer cannot tell the difference, but a human can do so easily. Another problem is that the relationship between the currently unknown languages and known languages is not established, and it is unknown if such a connection exists. Yet another problem is that in the unknown scripts, it might not be so easy to make out letters, diacritics and punctuation, or where one letter ends and another one begins.



Credit: SRK Branavan/MIT

A clay tablet with the Ugaritic script

For the researchers, the machine learning system is at least a powerful tool to help humans decipher languages, and considerably accelerate the process. Finding out Ugaritic took decades, and was the result of some serendipitous discoveries, such as an axe that said “axe” in Ugaritic. Every time a lost language has been deciphered, it is a process that takes decades. It requires an encyclopedic knowledge of the language, a lot of manual effort, and a good bit of luck. For these reasons, deciphering an unknown language is considered by scientists in general and linguists in particular to be a rare achievement.

In 2020, Barzilay and colleagues improved it by removing the requirement of starting off with a closely

related language. The system can now identify relationships with known languages on its own, using information from the clustering and ordering of consonants and vowels in the unknown language, and comparing them with many other known languages. For a lot of time, linguistics were of the opinion that Iberian is closely related to Basque, but recent researchers suggested that they were not. Some researchers even claimed that Iberian is not related to any known language. This remains a matter of contention. The system built by MIT researchers also did not find a deep relationship between the two languages. Again, the system was tested by using it to translate two scripts that have already been deciphered, Ugaritic and Linear B.

The system is based on historical study of linguistics, which is observing how languages change over time. In all the languages, the drifts in the sounds are similar. For example, the b sound can change to the p sound in the word, but it is unlikely to change into the k sound because of how clumsy the pronunciation would be.

Currently, the system can produce mappings of words in an unknown language to a known language. The researchers hope to improve the system in such a way, that they can derive the meaning of the word without actually reading it. What this means is that the content of the word can be understood, even if another language does not have the same word. This would involve entity recognition approaches that are already used in NLP technologies for known languages. By looking at the content and the context, it might be possible to identify words that represent people, locations or times. Knowing the historical and geographical context for the lost language can further guide this process towards a successful translation. 